Thermal effects

Solid: Has a fixed shape and a fixed volume. Particles are held closely together by strong forces of attraction called bonds. They vibrates backwards and forwards but cannot change their position.

Liquid: Fixed volume but can flow to fill any shape. Particles are close together and attract each other. But they vibrate vigorously that the attractions cannot hold them in a fixed position, so they can move past each other.

Gas: No fixed volume or shape and can quickly fill any space available. Particles are well spaced out and virtually free of any attractions. They move at high speed colliding with each other and the walls of their container

$$\text{Solid} \underset{\text{Freezing}}{\overset{\text{Melting}}{\rightleftharpoons}} \text{Liquid} \underset{\text{Condensation}}{\overset{\text{Boiling}}{\rightleftharpoons}} \text{Gas}$$

Brownian motion: Random movement of particles over short distances in short distances.

Internal energy: The total Kinetic and potential energy of all atoms or molecules in a material.

* The hotter a material is, the faster its particles move, and the more internal energy it has.

→ If a hot material is in contact with a cold one, the hot one cools down and loses internal energy, while the cold one heats up and gains internal energy. The energy transferred is known as heat.

* Thermal energy is used for heat and internal energy.

$$\text{Kelvin} \underset{+273}{\overset{-273}{\rightleftharpoons}} \text{Celius}$$

0 Kelvin = 273 celcius

Heat	Temperature
- Thermal energy	- The indication in the direction of heat movement
- Measured in Joules (J)	- Measured in kelvin (K) or celcius (°C),
- Cannot be negative	- Can be negative
- No instruments to measure	- Measured using a thermometer

Thermal expansion: When an object's volume increases due to being heated.

* When a steel bar is heated, particles speed up, and their vibrations take up more space, so the bar expands slightly in all directions. If the temperature falls, the reverse happens and the material contracts.

→ Gaps are left at the ends of bridges to allow for expansion

→ Overhead cables are left slack when suspended from poles or pylons to allow for contraction on cold days.

* Gases expand much more than liquids and solids. It depends on the strength of attraction between the particles. Attractions are very strong is solids, if the temperature rises and particles move faster, this has little effect on their speration because they are so tightly held together. In liquids, attractions are weaker, so the expansion is greater. In gases, attractions are extremely weak, so the expansion is much more greater.

Conduction: When a bar is heated, thermal energy is transferred from the hot end to the cold end as the faster particles pass on their extra motion to particles all around the bar.

More thermal energy is transferred if:
1- The temperature difference across the bar ends is increased.
2- The cross-sectional area of the bar is increased
3- The length of the bar is reduced

Conductors: Materials that are able to transfer heat efficiently
- Metals - Graphite - Silicon
Insulators: Poor conductors: Materials that transfers heat poorly.
- Non-metals - Materials that trap air - Water.

To reduce heat losses from a house:
- Plastic foam lagging around the hot water storage.
- Glasswool or mineral wool insulation in the loft
- Wall cavity filled with plastic foam, beads, or mineral wool
- Double glazed windows: Two sheets of glass with air between them.

* When a material is heated, the particles move faster, push on neighbouring particles and speed them up too.
* When a metal is heated, its free electrons speed up. As they move randomly within the metal, they collide with atoms and make them move faster.

Convection: Happens in fluids: As temperature increases, fluid expands, volume increases, density decreases, fluid becomes lighter, so particles of the fluid rise up (float)

During day time, in hot sunshine, the land heats up more quickly than the sea, because sand has lower heat capacity than water. Warm air rises above the land, as it is displaced by cooler air moving in from the sea.

During night time, the reverse happens. The sea stays warmer than the land, which cools down quickly. Warmer air now rises from the sea, as it is displaced by cooler air moving out from the land

Radiation: Infrared radiation and that all objects emit this radiation.
→ Thermal energy transfer by thermal radiation does not require a medium
→ For an object to be at a constant temperature: it needs to transfer energy away from the object at the same rate that it recieves energy.
→ If an object absorbs thermal radiation faster than it radiates, it heats up. If it radiates faster than it absorbs, it cools down.

Emitters	Best	———	Worst
	Black	White	Silver
Absorbers	Best	———	Worst
Reflectors	Worst	———	Best

* To compare emitters, a metal cube is filled with boiling water, which heats the surfaces to the same temperatures, one side is white, while the other side is matt black. The thermal radiation detector is placed in turn at the same distance from each surface, and the readings are compared.
* To compare absorbers, a matt black metal plate and a white metal plate are placed at the same distance on either side of a radiant heater. To find out which surface absorbs thermal radiation most rapidly, the rises in temperatures are compared.

To make a liquid evaporate more quickly

1- Increase the temperature

2- Increase the surface area

3- Blow air across the surface

Boiling	Evaporation
- Occurs at a particular temperature	- Occurs at a range of temperatures
- Relatively slow	- Relatively fast
- Takes place throughout the liquid	- Takes place at the surface
- Bubbles are formed	- No bubbles
- Temperature remains constant	- Temperature changes
- External thermal energy resource is required	- External thermal energy resource is not required

Specific heat capacity: The energy required per unit mass per unit temperature increase

$$C = \frac{Q}{m \times \Delta T}$$

Q → Thermal energy (J)

ΔT → Temperature (°C or K)

m → mass (Kg)

C → J/Kg°C